

A Coherent 1 μm Transmission System using TFLN Modulator, Gain-Flattened YDFAs and Low-Loss Hollow-Core Fibre

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Abstract We demonstrate the first full coherent 1 μm hollow-core fibre transmission system using new DNANF with lowest reported 1 μm loss, thin-film lithium-niobate modulator and custom gain-flattened YDFAs to deliver GMI-estimated 53.3 Tb/s over 106.1 km showing a path new spectral window adoption. ©2026 The Author(s)

Introduction

Hollow-core fibres (HCFs) have been widely explored as physical transmission channel for next generation optical networks due to several attractive features including low latency, low-loss, low nonlinearity, low-and-tunable chromatic dispersion and low back-scattering, which allow for high launch powers, long span lengths and bi-directional transmission [1, 2]. Despite many impressive system demonstrations [3-7], gas absorption has been shown to limit long-distance C-band transmission [5, 7] and to even make metro-scale distances challenging in the L-band [8]. However, the low-loss window of anti-resonant (AR)-HCFs is not fixed by material absorption, rather by the geometric design of the microstructure [1, 2]. This presents an opportunity to both move away from gas absorption limited spectral windows and to utilise new amplifier technology where bandwidth and power advantages may be exploited. Fig. 1(a) shows the relative absorption of H₂O and all gases from 1 to 2 μm at 20°C from the HITRAN molecular spectroscopic database [9].

The area of spectrum above 1 μm wavelengths, referred to as the 1060 nm, T- or Y-band is of interest due to low gas absorption over 10s THz, coupled with the availability of the ytterbium-doped fibre amplifier (YDFA). The YDFA is an established, high-power and low-noise amplifier, used widely in industrial applications and satellite communications, with similar pump diodes to EDFAs, but yielding potentially better optical-to-optical efficiency due to the reduced quantum defect due to the closer proximity between the pump and signal bands [10, 11]. The most common 1030 nm to 1085 nm transmission window can provide high gain over a bandwidth of 13.5 THz, comfortably exceeding the 9-10 THz of combined C+L-band Erbium (E)-DFAs. Moreover, low-noise amplification bandwidths up to 21.9 THz [12] have been demonstrated using more complex YDFA configurations.

Previously, directly modulated VCSELs and high-bandwidth modulators [13-15] operating in 1 μm have been reported. HCFs have also been showed to be able to achieve low losses in this spectral window [16-18]. For data transmission, intensity modulation at 1 μm was explored in short HCFs with low spectral efficiency [19, 20]. Most recently, a 7.12-Tb/s transmission of 31×3-mode, Kramers-Kronig signals in SMF (G.652) with 15 nm bandwidth in the 1 μm region was reported [21], as summarized in Fig. 1(b).

Here, we demonstrate a full 1 μm -band transmission system comprising novel TFLN modulator, telecom grade gain flattened YDFAs and new custom-designed HCF with potential for >50 THz of low loss bandwidth and the lowest loss of any 1 μm -band transmitting fibre. Over a 106.1-km fibre span, we transmit 244×32GBd, 50-GHz-spaced dual-polarization probabilistically-shaped 16-ary quadrature amplitude modulation (DP PS-16QAM) data channels from around 1030 nm to 1075 nm. A total system data-rate of ~53.3 Tb/s is achieved. These results show that low-loss AR-HCFs can become the physical transmission medium needed to unlock the data-carrying potential of the YDFA to enable high-capacity single amplifier systems with more bandwidth than combined C+L-band EDFAs.

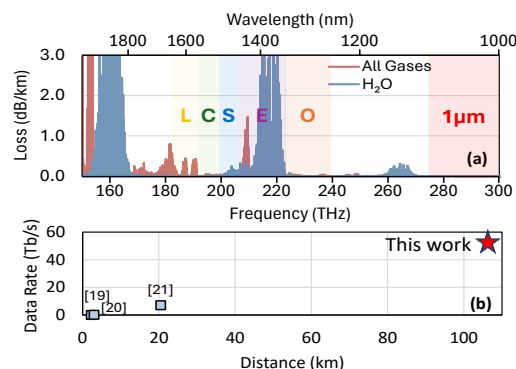


Fig. 1 (a): Absorption of common gas & water 150-300 THz (~1-2 μm) and (b) Recent 1 μm transmission demos.

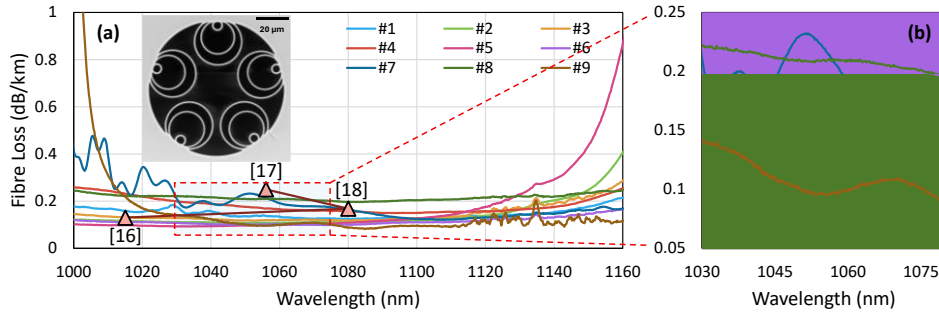


Fig. 2: (a) Measures losses of the used 1 μm fibres and prior loss records in the 1 μm region, (b) zoom-in plot of the transmission window in this work. Inset: fibre cross section image.

Low-Loss 1 μm Hollow-Core Fibre

The transmission fibres were five-tube double nested anti-resonant nodeless fibres (DNANFs) similar to those reported previously in the C-band [2]. This design consists of five sets of non-contacting nested capillaries surrounding the core region, with core and tube diameters informed by modelling and chosen to optimize the loss whilst imposing high attenuation on the higher-order modes with the anti-resonant windows approximately bounded by the ARROW-defined membrane resonances occurring at $\lambda_m = 2t / m\sqrt{n_g^2 - 1}$ where t is the tube wall thickness, m the resonance order, and n_g the refractive index of glass. For this work, we opted for operation in the second anti-resonant window at 1 μm , making the membrane thickness ~ 750 nm.

The fibres were fabricated using the stack, fuse, and draw approach. Their average core diameters are between 25 and 26 μm , with the gap between adjacent outer capillaries being ~ 4 μm . Fig. 2 shows representative attenuation spectra of the fibre samples used in later transmission experiment, which were measured by cutback from the full band length to about 20 m. A minimum loss of ~ 0.08 dB/km was recorded in several samples between 1.0 and 1.1 μm with the loss remaining below 0.2 dB/km from ~ 930 nm to 1130 nm, corresponding to a usable bandwidth of ~ 57 THz. Fig 2(a) shows that all fabricated samples share at least 30 THz low-loss bandwidth covering the primary and extended YDFA transmission window.

The fibre span comprises 9 fibre sections with

length between ~ 7.4 and 18.8 km with a total length of around 106.1 km. The splice loss was estimated to be around 0.15 dB per splice and the total span loss measured to be ~ 19.6 dB at 1060nm, giving an effective span attenuation of ~ 0.185 dB/km. The measured intermodal interference remains below -55 dB/km, confirming that the span supports high-order coherent modulation without penalty from modal beating.

Experimental Setup

The experimental setup is shown in Fig. 3(a). A test channel was based on light from an external-cavity tunable laser with <100 kHz linewidth. The 1064-nm TFLN single-polarization IQ Mach-Zehnder modulator (IQM) combines high efficiency with ultrabroad bandwidth using compact 2.9-mm traveling-wave electrodes [15]. It exhibits $V_\pi=4.2$ V and low $V_\pi L=1.1$ Vcm, approximately twice as efficient as typical 1550-nm TFLN modulators. The measured 3-dB electro-optic bandwidth exceeded the 110-GHz measurement limit. The device was packaged with glued optical fibers, and DC connectors. Only a low bandwidth Si RF interposer was available for feeding the input signal which limited the packaged RF bandwidth to below 20 GHz.

The modulator was driven by a digital-to-analogue converter at 256 GSa/s with root-raised-cosine (RRC) shaped DP PS-16QAM symbols at 32 GBd. The roll-off factor was 0.01, and entropy was set as 3.8 bit/symbol. A polarization emulation stage using a pair of polarization beam splitter/combiner (PBS/PBC) with delay line was

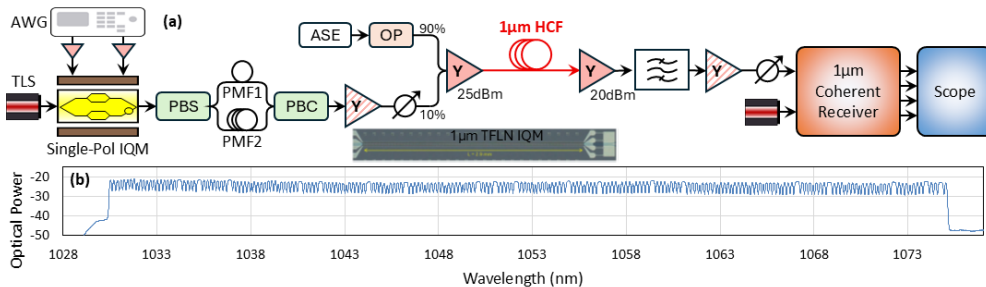


Fig. 3: (a) Experimental setup of the 1 μm transmission system over HCF, and (b) the WDM optical spectra at the transmitter with 4-GHz resolution. Inset: microscope picture of a single arm of the 1 μm TFLN IQM.

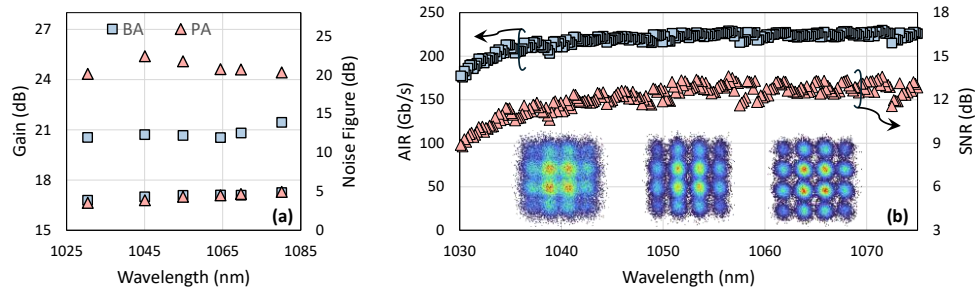


Fig. 4: (a) Gain and NF for BA and PA with +5 dBm and -5 dBm input powers and **(b)** SNR and AIR of received channels.

adopted. The test channel was stepped across the transmission band from around 1030 nm to 1075 nm with the remaining launch spectrum filled with shaped amplified spontaneous emission (ASE) noise emulating dummy channels. We note that although the YDFAs and fibre allowed for a wider transmission band, the range was limited by the tunable lasers. The ASE was shaped using a liquid crystal-on-silicon (LCOS) based optical processor (OP) after a high-power ASE source. The test and dummy channels were combined in a 10-dB coupler fed to a booster YDFA. The transmitter spectra are shown in Fig. 3(b) and its resolution is limited by the OSA.

All YDFAs were based on a 2-stage design with mid-span gain-flattening filter. The launch power was set to 25 dBm for the DWDM booster YDFA (BA), equivalent to just over 1 dBm per channel, and the combined signal was then directed to the transmission span using a free-space mode-field adaptor. After fibre transmission, a low-noise YDFA optimised for low power DWDM signal (PA) and single-channel optimised YDFA were placed either side of a tunable filter used to select the test-channel for reception. Fig. 4(a) shows the gain and noise figure performance of the BA and PA at 5 dBm and -5 dBm input powers, respectively. A bulk coherent receiver consisted of a custom 90° optical hybrid and wide-band balanced photodetectors (BPDs) operating between 1000 to 1700 nm, whose outputs are captured by a real-time oscilloscope at 80 GSa/s for offline processing.

The offline processing includes front-end impairments corrections, resampling to 2 samples/symbol, dispersion compensation, RRC matched filtering, frame synchronisation, pilot-aided fractionally-spaced adaptive MIMO equaliser for polarisation demultiplexing, pilot-aided carrier recovery, and decision-directed equalisation. Signal-to-noise ratio (SNR) and achievable information rate (AIR) estimated by generalised mutual information (GMI) [22] were evaluated, where the 3.125% pilot was excluded.

Experimental Results

Fig. 4(b) shows the received AIR and SNR as a

function of wavelength for all the received 244 channels. The AIR and SNR are consistently around 220 Gb/s and 12.5 dB, respectively, for wavelengths above ~1050 nm. Below that, both start to decay, reaching ~180 Gb/s and ~9 dB near 1030 nm. The decay in AIR and SNR at the shorter wavelengths is attributed to the roll off in responsivity of the BPDs and reduced stability of the modulator bias circuitry, although the YDFAs' gain is still high. The total system GMI-estimated data-rate for all 244 received channels was ~53.3 Tb/s from the ~12.15-THz utilised bandwidth, and no evidence of any impairment from gas absorption is observed.

We note there is considerable scope for improved performance with improvements to the modulator packaging allowing access to the full device bandwidth for more spectrally efficient modulation. Further, the highest wavelength channel of ~1075 nm was limited by the tuning range of the available tunable lasers rather than amplifiers or fibres. It is also expected that much longer unrepeatable lengths are possible with only 25 dBm output power required for the fiber length available, despite availability of flattened amplifiers with up to 36 dBm output power and literature reports of 100W amplifiers [11]. These opportunities show strong potential for building on this demonstration and encouraging development of a new eco-system in this spectral region.

Conclusions

We demonstrated the first full coherent metro-scale system utilizing a record-low-loss 1 μ m HCF in combination with gain flattened YDFAs and TFLN modulator. Over a 106.1-km span, the system transmitted 244×32 GBd DP-PS-16QAM channels across around 45 nm/12.15 THz, limited by the range of tunable lasers. The aggregate GMI-estimated throughput was ~53.3 Tbit/s with no evidence on impairment from gas absorption and potential for extended reach with higher output power amplifiers. The result shows that low-loss 1 μ m HCFs can enable single amplifier systems with bandwidth greater than combined C+L-band and unlock the data-carrying potential YDFAs for next generation optical networks.

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